

Research Methodology
Question bank for external examination

Module 1

UNIT 1

1. Explain the role of business research & information system.
2. Distinguish between information system & knowledge management.
3. Explain in detail qualitative research tools.
4. Explain problem definition & secondary data research.
5. Write a short note on:-
 - i) Research
 - ii) Methodology
 - iii) Research Methodology

UNIT 2

1. Explain experimental research.
2. Explain in detail survey research communication with respondents.
3. Explain in detail observation methods.
4. Explain steps in conducting observation research.
5. Explain Market Research, Employee Research, Education Research & Healthcare Research.

Module 2

UNIT 3

1. Explain levels of scale measurement.
2. Explain in detail methods of attitude measurement.
3. Explain types of sampling design.
4. Explain factor affecting sample size.
5. Explain questionnaire design.

UNIT 4

1. What is basic data analysis? Explain in detail.
2. Explain univariate statistical analysis.
3. Explain multivariate statistical analysis.
4. Distinguish between Univariate & Bivariate analysis.
5. Explain bivariate statistical analysis.

Research Methodology

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Q1) Role of Business Research & Information System

(A) → Business research plays a crucial role in RM by providing a systematic & scientific approach to solving business problems.

— The following are the key roles of business research:

- Helps in identifying real business problems
- Clearly defines research objectives
- Converts problems into researchable questions
- Helps in data analysis & interpretation

(B) → Information systems support business research by providing accurate & timely information.

— The following are the key roles of information systems:

- collects raw data from various sources
- Processes data into meaningful information
- Store large amounts of data efficiently
- Provides reports for planning & control.

Q.2) Distinguish between Information System & Knowledge Management

→ Info. System

Knowledge Management.

① Focus on data & Info.

Focus on knowledge & experience

② Technology-oriented

People-oriented

③ Uses hardware & Software

Uses learning & Sharing

④ Stores data in DB

Stores explicit

⑤ Processes raw data

Converts knowledge into action

⑥ Deals mainly with structured data

Deals with Structured & Unstructured Knowledge

⑦ Provides reports & summaries

Share best practices & expertise

⑧ Used for operation decisions

Used for strategic & long term decisions

⑨ Eg: MIS, TPS, DSS

Eg: Training programs, Knowledge portals.

Q.3) Explain in details Qualitative Research Tools

- Qualitative research tools are used to collect non-numerical data and to understand opinions, attitudes, perceptions, experiences & behavior of individuals or groups.
- These tools help the researcher gain in-dept & detailed insights into a research problem rather than focusing on numbers or statistics.

— The following are the main qualitative research tools:

(I) Interviews :

- Direct interaction between researcher & respondent.
- Can be face to face, telephonic or online
- May be structured, semi-structured or unstructured

(II) Observation :

- Data is collected by directly watching behavior or events.
- Researcher does not depend on respondent's answer.
- Provides realtime & authentic data.

(III) Document & Record Analysis:

- Uses existing documents, reports, & records.
- Helps understand background & historical info.

(IV) Case Study (V) Group Discussion

Q.4) Explain in detail Observation Method.

→ Observation method is a research technique in which the researcher directly watches & records the behavior, actions, or events of individuals in a systematic manner.

• This method is commonly used in Social, business, and market research to collect real time & First hand ^{data}

- Types of Observation Method:

(i) Structured Observation — Uses predefined checklists & Categories

(ii) Unstructured Observation — No fixed format

- Advantages of Observation Method:

- Provides first-hand & real time data
- Captures actual behavior
- Useful when respondents cannot explain their actions.

Q5) Write a Short notes on: . . .

(a) Research:-

- Research is a systematic & scientific process used to discover new knowledge & solve problems.
- It involves collecting, analyzing & interpreting data in an organized manner.
- Research helps in finding facts, verifying existing knowledge & developing new ideas.

(b) Methodology:-

- Methodology refers to the procedure or approach used to conduct research. It explains
- It explains how the research is carried out & which methods & techniques are used.
- Methodology ensures that research is conducted in a proper & organized way.

(c) Research Methodology:-

- Research methodology is the overall framework or blueprint of research.
- It includes the research design, data collection methods, sampling techniques, & data analysis procedures.
- Research meth. guides the entire research process from beginning to end.

Q.6) Explain problem definition & Secondary Data Research.

→ Problem definition is the process of clearly identifying & stating the research problem that needs to be studied.

• It is the first & most imp step in the research process.

— The following are the ^{point of} key problem's ~~distinction~~ ^{distinction}:

- Clearly identifies what problem is to be studied
- Helps in setting clear research objectives
- Distinguishes between symptoms & the actual problem
- Saves time cost & effort.

→ Secondary data research involves the use of data already collected by others for purposes other than the current research.

• It is cost effective & time saving method of research.

— Key Points :

- Uses existing data sources
- Data may be internal or external
- Save time & reduces cost of data collection

Module - 2

Answer

(UNIT-3)

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(Q.) Explain levels of Scale measurement.

→ levels of Scale measurement refer to the different ways in which data can be classified, ordered & measured in research.

• There are four levels of measurements :

(i) Nominal Scale :

- Simplest form of measurement
- Data are classified into categories only
- No ranking or order

Eg : Gender, Religion, blood group, Nationality

(ii) Ordinal Scale :

- Data are classified & ordered
- Ranking is possible
- Difference between ranks is not measurable.

Eg : Class rank, Grades, Socio-economic Status

(iii) Interval Scale :

- Ordered data with equal intervals
- No true zero point

Eg : Temperature ($^{\circ}\text{C}$, $^{\circ}\text{F}$), Calendar year, IQ

(iv) Ratio Scale :

- Highest level of measurement
- Has true zero point • All mathematical operations possible

Eg : Age, Height, Income, Weight.

Q) Explain in detail methods of attitude measurement.

→ Attitude measurement means finding out what people think, feel, or believe about a particular topic, product, service or idea.

• Attitude has three components (ABC Model):

(I) Affective: Feelings (like/dislike)

(II) Behavioral: How a person behaves

(III) Cognitive: Belief or opinions

— To measure attitude properly, several methods are ^{used}

(V) Likert Scale:

• The most commonly used method

• Options usually range from Strongly Agree to Strongly Disagree.

• Example "I am satisfied with my job"

Strongly Agree → Strongly disagree

Thurstone

(II) Thurstone Scale: (Equal Appearing Interval Scale)

• Experts / judges assign a score / weight to each statement.

• Respondents select the statements they agree with

(III) Semantic Differential Scale:

- Uses bipolar adjectives (two opposite words)
 - Respondents choose a position between the two
- Example: Good / bad, Useful / Useless

(IV) Guttman Scale (Cumulative Scale)

- Contains statements arranged from easy to difficult or from low to high intensity.
- If a person agrees with a high-intensity statement they are assumed to agree with all lower statements

Example:

#1. I like this product

#2. I recommend this product

#3. I always buy this product

If a respondent says "Yes" to #3, they are assumed to say "Yes" to #1 & #2 also

(V) Bogardus Social Distance Scale:

- Measures social distance or comfort level with different social groups.
- Common in sociology.

Example:

would you accept someone as:

Neighbours?

Co-worker?

Close-friend?

Less acceptance = more social distance.

Q. Explain types of Sampling Design.

→ A sampling design is the method used to select a sample from a population.

It is mainly divided into two types:

(A) Probability Sampling: —

— Every unit has a known & equal chance of being selected.

(i) Simple Random Sampling:

- Each member has an equal chance of selection
- Methods: lottery, random numbers.

(ii) Stratified Sampling:

- Population is divided into strata (group)
- Samples taken from each group for better representation

(iii) Systematic Sampling:

- Selecting every k^{th} member from a list
- eg, every 10th person.

(B) Non-Probability: —

Selection based on judgement; (unequal chances)

(i) Convenience Sampling: Selecting whoever is easiest to contact.

(ii) Judgment (Purposive) sampling: Researcher selects specific people based on judgement.

(III) Quota Sampling: Fixed quotas for categories (like gender quota)

(IV) Showball Sampling: Existing respondents refers new respondents.

Q) Explain Factor affecting Sample Size.

→ Sample Size means the number of people/items selected from the population

- The correct sample size is imp. for accurate results
- The main factors that affect sample size are:

(I) Population Size:

- If the population is larger generally the sample size should be bigger.
- For small populations, a small sample may be enough.

(II) Nature of population:

- If the population is similar (homogeneous), a small sample is enough.
- If the population is different (heterogeneous), a bigger sample is needed.

(III) Level of Accuracy Needed

- Higher accuracy → larger sample
- Lower accuracy → smaller sample

(IV) Confidence Level :

- Common levels : 90% , 95% , 99%.
- Higher Confidence level requires a larger sample.

(V) Variability in Data :

- If data values vary a lot, a bigger sample is needed.

(VI) Time & Cost :

- Limited time, money or resources may reduce the sample size.

(VII) Type of Analysis :

- Simple analysis needs small sample.
- Advance statistical analysis requires a larger sample.

Q.) Explain Questionnaire design.

- • A Questionnaire is a structured set of written questions used to collect information from respondents.
- Questionnaire design means planning, writing & arranging questions in a proper way so that the data collected is accurate & easy to analyze.

- Types of Questions in a Questionnaire

(a) Open-Ended Questions :

- Respondents can answer in their own words.
- Eg - "What problems do you face at work?"

(i) Closed-Ended Questions:

- Respondents choose from given options.
- Ex - "Are you satisfied with your job?" (Yes/No)

(ii) Dichotomous Questions:

- Only two options: Yes/No, True/False

(iii) Multiple Choice Questions:

- More than two choices are given.

(iv) Scaling Questions:

- Used to measure the intensity of opinion
- Ex - Strongly Agree $\rightarrow$ Strongly Disagree

(v)

Q. What is basic data analysis?

→ Basic data analysis means organizing, summarizing, & interpreting collected data so that meaningful conclusion can be drawn.

- It is the process of converting raw data to useful information.

Steps in Basic Data Analysis

(i) Editing:

- Checking data for errors, missing values & inconsistencies.
- Types of Editing (A) Field Editing (B) Central Ed

(ii) Coding:

- Assigning numbers or symbols to answers.
- Types of coding (A) precoding (B) post coding

(iii) Classification:

- Grouping data into categories (e.g. age group, income group)
- Types: Qualitative, Quantitative, Geographical

(iv) Tabulation:

- Presenting classified data in table form
- Helps to compare values easily.

(v) Diagrammatic & Graphical Presentation:
 • Using bar charts, Pie chart, line graphs, histograms etc.. for better understanding.

(vi) Statistical Analysis:
 • Using mean, median, mode, Standard deviation for summarizing data.

Q.) Explain Univariate Statistical analysis

- • Univariate Statistical analysis mean analyze one variable at a time.
 • It helps to describe & summarize the basic characteristic of a single variable.

— Technique Used:

(i) Frequency Distribution — Shows how many times value occurs.

(ii) Measures of Central Tendency
 • Mean (average)
 • Median (middle value)
 • Mode (most frequent value)

(iii) Measures of Dispersion
 • Range
 • Variance
 • Standard Deviation

(IV) Graphical Representation

- Bar Chart
- Pie Chart
- Histogram

Example :

- | | |
|------------------|------------------|
| • Age of Student | • Monthly income |
| • Marks scored | |

Q.) Explain Multivariate Statistical Analysis

- Multivariate Statistical analysis involves analyzing three or more variable at the same time.
- It is used when researcher wants to study complex relationships among multiple factors.

Common Multivariate Techniques:

(i) Multiple Regression:

- Studies the effect of two or more independent variables on one dependent variable.
- Eg: $Sales = f(\text{price, Advertising, Income})$

(ii) Factor Analysis:

- Reduces many variables into a few meaningful factors.
- Eg: Customer satisfaction factors like Price, Quality, etc.

(iii) Cluster Analysis:

- Groups people or objects with similar characteristics.
- Eg - Market Segmentation.

- (iv) Discriminant Analysis -
• classifies items into predefined groups.
• Eg; loyal vs non-loyal customers.

- (v) Principal Component Analysis -
• Data reduction technique similar to factor analysis.

Q) Explain Bivariate Statistical Analysis

→ Bivariate Statistical analysis is the analysis of variables at the same time.

• The main purpose is to study the relationship, association or effect between two variables.

Common Techniques Used: -

- (i) Cross Tabulation: Shows the relationship between two categorical variables in the table.
- (ii) Correlation Analysis: Measures the degree & direction (positive/negative) of the relationship.
- (iii) Simple Regression Analysis: Shows how one variable (independent) affects another (dependent).
- (iv) Chi-Square Test: Used to test the association between two categorical variables.

(V) Scatter Diagram : A graph used to visually see how two variables are related.

Examples:

- Relation Ship between income & expenditure
- Relationship between advertising & Sales
- Relationship between education level & income

Q.) Distinguish between Univariate & Bivariate Analysis

	Univariate	Bivariate
No. of Variables	One variable only	Two variables
Purpose	To describe & summarize data	To study the relationship between variable
Nature	Descriptive	Analytical
Technique Used	Mean, Median, Mode,	Correlation, Regression, chi-square
Graph	Bar chart, Pie chart, Histogram	Scatter diagram
Data Complexity	Simple & easy	more complex
Outcome	Shows characteristics of one variable "What is the value?"	Show association between two variable. "Are the variables related?"
Types of COA	Distribution of a single variable	Direction & strength of relationship
Statistical focus	Variable	Studying income vs. expenditure.
Example	Studying age of students	